

No of Pages : 4

Course Code : 23MX12

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004
SEMESTER EXAMINATIONS, MARCH 2025

MCA Semester : 1

23MX12 STRUCTURED PROGRAMMING CONCEPT

Time : 3 Hours

Maximum Marks :100

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivisions (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.

1. a) Compare and contrast top down approach and bottom approach for structured programming.
- b) i) Discuss on various criteria for classification of programming languages. Based on the secriteria, fit the position of C Programming language.? (5)
- ii) Consider the following program. Write the output for every input shown in the table (5)

```
#include<stdio.h>
#include<conio.h>
int main() {
int x=0,n;
clrscr();
scanf("%d",&n);
if(n>8) {
if(n<9) x=1; }
else {
if(n>4)
if(n<6) x=2;
else x=3;
else
if(n>2) x=4;
else
x=5; }
printf("%d",x); }
```

Input	1	3	7	9
Output				

- c) Analyze the properties of good programming language in the perspective of programmer and language developer. Enumerate the reasons for the popularity of a programming language. Give simple examples in C programming for the above points.

2. a) Demonstrate a situation where switch case is better than if else ladder. Find the output of the program

```
#include<stdio.h>
#include<conio.h>
void main() {
    clrscr();
    switch printf("a"), printf("abc"), printf("ab")
    {
        case 1: switch(printf("first"))
            break;
        case 2: switch(printf("second"))
            break;
        default : switch(printf("third"))
            break;
    } }
```

- b) Analyze following aspects of structured programming language with C language and compare them.

- i) Scope and lifetime of variables.
- ii) Call by value and call by address
- iii) Iterative structure statements and its sequence control

- c) i) 1) Explain the various components of user defined function with example? Also distinguish iteration from recursion in software application development with suitable illustration. (8)

- 2) Develop a C function to perform the following task. Ramya has two numbers, her tuition teacher give a task to her. The task is that, she wants to find the sum of all digits appearing between those two numbers. (4)

Num1 = 8 and Num2 = 13

Output: 27

Explanation: (8 + 9 + 1 + 0 + 1 + 1 + 1 + 2 + 1 + 3)

Constraints

0 <= Number1 <=10000

Number1 <= Number2 <=10000

(OR)

- ii) Consider the C program below (with gaps to be filled in) to perform the following task:

Reads a positive integer k ($0 < k < 5$) and then prints all permutations (with repetitions allowed) of length k using the digits from 1 to 9. Note that since repetitions are allowed, there will be 9^k permutations for a given k . You are required to fill up the missing lines in the program.

Note: Interchanging i_2 , i_3 , and i_4 as loop control is okay as long as the correct permutations are printed

<pre> void main() { int k, count = 1, i1, i2, i3, i4; scanf("%d", &k); printf("k = %d\n", k); for (i1 = 1; i1 < 10; ++i1) if (k == 1) { printf("Permutation No %d = ", count++); printf("%d\n", i1); } else for (i2 =-----) if (k == 2) { printf("Permutation No %d = ", count++); printf(" %d, %d\n", -----); } else for (i3-----) if (k==3) { printf("Permutation No %d = ", count++); printf-----"); } else for (i4 -----) if (k==4) { ----- printf("Permutation No %d = ", count++); printf-----); } } </pre>	<----- <----- <----- <----- <----- <----- <-----
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3. a) Differentiate constant pointer and a pointer to a constant.
- b) Explain any five string functions used in C?
 Write a C program to implement the following. Given a string s , you have to replace all instances of
- 'AND' with '&'
 - 'QN' with '?'
- and print the resultant string.

Test case:

Consider $s = \text{'and_is_AND_but_and_is_not_AND_though_QN_is_?'}$

The output must be

'and_is_&_but_and_is_not_&_though_?_is_?'

- c) i) 1) Differentiate static and dynamic memory allocation and Discuss the functions used in dynamic memory allocation. (6)
- 2) Implement using dynamic memory allocation, pointer and pointer arithmetic operations to accept two matrices and store them in column major form and perform matrix concatenation of corresponding matrix elements.
 $X[i][j] = 3$ and $Y[i][j] = 7$ and the result $Z[i][j] = 37$ (6)
- (OR)
- ii) Write a structure to store the name, account number and balance of customers (more than 5) and store their information in a file.
- 1) Write a C program that opens the file and appends the given new customer records to the file.
- 2) Display the content of the file.
- 3) Write a function to print the names of all the customers having balance less than Rs.200.
4. a) Under what scenario, scripting languages are used for developing applications.
- b) Compare multi-paradigm languages with domain-specific languages, and also discuss when would you choose one over the other with typical use cases?
- c) Explain the concept of static and dynamic libraries using C programming.

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FD/JU

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Course Code : 23MX12

Roll No:

(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004**SEMESTER EXAMINATIONS, JUNE 2024****MCA Semester : 1****23MX12 STRUCTURED PROGRAMMING CONCEPTS****Time : 3 Hours****Maximum Marks : 100****INSTRUCTIONS:**

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivisions (a) carries 3 marks each, subdivision (b) carries 7 marks each and subdivision (c) carries 15 marks each.

1. a) Compare and contrast Top-Down approach and Bottom-Up Approach for structured programming?
- b) Discuss Why C is considered as one of the best programming languages for developing System software and tools.
- c) Compare and Contrast properties of Programming language with respect to programmer and Language developer with illustrations from 'C' language
2. a) Consider a C program that takes a character input. If the input character is an uppercase character between 'A' and 'Z', then the program prints the character in lowercase; otherwise, it simply prints the input character. Fill up the missing code below to complete this program.

```
#include int main() {
    char c;
    printf("\nEnter the character:");
    scanf(" %c",&c);
    if((-----))
    { -----; }
    printf("\nThe output character is %c",c); }
```

- b) i) Complete the following code snippet is from a program that takes as input a stream of integers until the number 0 is entered. It then prints the total number of even (zero excluded) and odd integers that were given as input [3]

```
#include <stdio.h>
int main() {
    int odd_count=0, even_count=0, num;
    printf("\n Enter integers (input 0 to terminate):");
    scanf("%d", &num);
    while(_____)
    {
        if(_____)
            even_count++;
        else
            odd_count++;
        _____;
    }
    printf("\n The number of even numbers entered is %d.", even_count);
    printf("\n The number of odd numbers entered is %d.", odd_count);
    return 0; }
```

- ii) Discuss two types of Iterative structure statements with their syntax, semantics and sequence controls in programming language 'C' with illustrations [4]
- c) i) Explain the following constructs with illustrations from C language, when to use and when not use while programming by discussing the principles behind the constructs. [5 x 3 =15]

1. Expressions vs. statements
2. Literals, constants and variables
3. Header files and Library files
4. printf () vs. scanf ().
5. Sequence point and Expression Evaluation

[OR]

- ii) 1) Write down the output for the following programs. If you think that there will be a compilation error or a runtime error, mention that as the output. [4 x 1.5 =6]

a. What will be the output of the following program? <pre>#include<stdio.h> int main(){ int a=10; a+=a*a%2; a%=2; if((a=0)) printf("Correct"); else printf("Wrong"); return 0; }</pre>	b. Find out the output of the following program. <pre>#include<stdio.h> int main(){ int i, j; for (i = 0; i <= 2; i++) { for (j = 0; j <= 2;) { printf("%d ",j); break; } printf("%d ",i); } return 0; }</pre>
c. What would be the output of the following code? <pre>#include<stdio.h> int main() { int j=(printf("TEN="),10); printf("%d",j); return 0; }</pre>	d. What would be the output of the following code? <pre>#include<stdio.h> #define CUBE(X) X*X*X int main() { int j=(printf(" CubE(2+2)="),CUBE(2+2)); printf("%d",j); return 0; }</pre>

- 2) Write a C program to find the saddle point from the given matrix of size M x N. Note The saddle point is a value which is minimum in the row as well as maximum in the column [9]

3. a) What are the outputs of this code?

```
struct _st {
    int x,y;
    struct _st *lnk1,*lnk2;
} a,b,c, *p;

a.x=b.y=10;    a.y=b.x=15;
c.x=a.x+b.x;   c.y=a.x+b.y;
a.lnk1=&b; b.lnk1=&c; a.lnk2=&c; b.lnk2=&a;
c.lnk1=c.lnk2=NULL; p=&b;
printf("val1=%d val2=%d \n",p->lnk1->x,p->lnk2->y);
printf("val3=%d val4=%d \n",p->lnk2->lnk1->x, p->lnk2->lnk1->y);
```

- b) Find the output and describe the output generated by the following program [7]

```
# include <stdio.h>
int funct1 (int );
int funct2(int);
main() {
    int a=0,b=1,count;
    for ( count =1; count <= 5 ; ++count)
    {   b+=funct1(a++) + funct2(a++);
        printf("%d",b++);
    }
}
int funct1( int a)
{
    static int b;
    b= funct2(a++);
    return b++;
}
int funct2(int a)
{
    static int b= 1;
    b+=1;
    return(b++ + a++);
}
```

- c) i) Discuss the following aspects of structured programming **with** illustrations from C and compare them

- 1) Call by Value and Call by Reference [5]
- 2) Storage Classes [5]
- 3) Arrays and pointers [5]

[OR]

- ii) Write a program that accepts two matrices and store them in column-major form and perform matrix concatenation of corresponding matrix elements.

$A[i][j] = 5$ and $B[i][j] = 6$ and the result is $c[i][j] = 56$.

Use dynamic memory, Pointer and Pointer Arithmetic operations

4. a) Compare and contrast the use of Scripting and Markup languages in Web Client and Server side applications development?
- b) Discuss and Compare features and flexibilities of System programming Languages and Script languages in details.
- c) Explain the features of C language "Static and Dynamic libraries and Low level programming " with real-life applications. [8+7]

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FD/RL

No of Pages :4

Course Code : 23MX12

Roll No:
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PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2024

MCA Semester: 1

23MX12 STRUCTURED PROGRAMMING CONCEPTS

Time : 3 Hours

Maximum Marks :100

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.

2. Course Outcome :

Qn.1	CO1.
------	------

Qn.2	CO2.
------	------

Qn.3	CO3.
------	------

Qn.4	CO4
------	-----

Table

COs	At the end of learning this course, students will be able to
CO1	Choose appropriate Programming Paradigm, language and environments for problem solving and program development.
CO2	Analyse language constructs of imperative paradigm for program development.
CO3	Organize the software applications by analyzing and using proper structured solutions in Coding and data structures.
CO4	Develop programs using programming constructs to solve problems with emphasis on best practices like SCM, Patterns and Frameworks opted

Question

(BTL)

1. a) Compare and contrast top down approach and bottom approach for structured programming. (L4) (3)
- b) i) Demonstrate with suitable illustration passing arguments at runtime? (5) (L3)
- ii) Analyze the code given below and write atleast 5 test cases. (5)
- ```
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char *argv[]) {
 if (argc != 3) {
 printf("Insufficient Arguments!");
 return 1;
 }
 int num1 = atoi(argv[1]);
 int num2 = atoi(argv[2]);
 int sum = num1 + num2;
 float quotient = (num2 != 0) ? (float)num1 / num2 : 0.0f;
 printf("Sum: %d\n", sum);
 printf("Quotient: %.2f\n", quotient);
}
```

```

 return 0;
}

```

- c) Discuss various characteristic features for classification of programming languages. Identify the position of C programming language in classification chart. For what type of applications C language shall be used for software development? (L3) (12)
2. a) Demonstrate a situation to use a while statement where a for loop would not be appropriate. (L3) (3)

Convert the following for-loop into an equivalent while-loop

```
for(i = 2; i <= sqrt(n); i += 3)
```

```
printf("%d ", i);
```

- b) i) Analyze the scope and lifetime of the following variables, providing examples for each (a) External variable (b) Static variable (c) Automatic variable (d) Register variable. (L4) (6)
- ii) Trace and Write the output for the given code. (4)

```
#include <stdio.h>
```

```
void call();
```

```
int main()
```

```
{
```

```
 call();
```

```
 call();
```

```
}
```

```
void call()
```

```
{
```

```
 inti = 11;
```

```
 int k=0;
```

```
 printf("%d ", i);
```

```
 static int j = 12;
```

```
 j = j + 1;
```

```
 k = ++j;
```

```
 printf("%d%d\n", j,k);
```

```
}
```

- c) i) Discuss all decision making and iterative statements with their syntax, semantics and sequence controls in C programming language with suitable illustrations. (8+4)

Trace and Interpret the output of the following program. (L4)

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
 inti = 1, j = 1, k = 1, count = 0;
```

```
 while (i < 2) {
```

```
 for(; j < 4; j += k)
```

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```

do {
 ++count;
 k += i;
} while (k < 8);
i += j;
}
printf("Loop Indices: %d %d %d\n", i, j, k);
printf("Number of iterations = %d\n", count);
return 0;
}

```

(OR)

- ii) Discuss the various components and types of user defined function? (L4)

What could happen if a recursive function lacks a proper base case or if the base case is never reached? Give a suitable example. Analyze the code step-by-step and write the output. (6+6)

```

#include <stdio.h>
void trickyRecursion(int *arr, int n) {
 if (n == 0) {
 return;
 }
 *arr += n;
 trickyRecursion(arr + 1, n - 1);
 *arr += n;
}
int main() {
 int arr[] = {1, 2, 3, 4, 5};
 int size = 5;
 trickyRecursion(arr, size);
 for (int i = 0; i < size; i++) {
 printf("%d ", arr[i]);
 }
 return 0;
}

```

3. a) What do you mean by enumeration in C and how does it help in code readability and maintainability. Trace and find the output of the code snippet below. (L2) (3)

```

#include <stdio.h>
enum designFlags{ CROP = 1, ROTATE = 2, SAVE = 4};
int main() {
 int myExample = ROTATE | SAVE | CROP;
 printf("%d", myExample);
}

```

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```

return 0;
}

```

- b) Analyze the different ways of initializing string? Explain any five string functions used in C? (L4) (6+4)

Write a C program to implement the following You are given two strings s1 and s2 of equal length. A string swap is an operation where you choose two indices in a string (not necessarily different) and swap the characters at these indices. Return true if it is possible to make string S<sub>1</sub> same as string S<sub>2</sub> by performing at most one string swap Otherwise, return false. (L6)

- c) i) 1) Differentiate static and dynamic memory allocation and Discuss the functions used in dynamic memory allocation. (L2) (6)

2) Write a C program that dynamically allocates memory for a 2D array of strings, takes input from the user, and uses pointer arithmetic to identify and display the minimum element in the array. (L6) (6)

(OR)

- ii) Explain the difference between the 'r' and 'r+' modes when opening a file in C. Provide an example for each mode to demonstrate their use. (L2)

Write C program that accomplishes the following. Consider the structure

```

Struct item{
 int id;
 char name[20];
 int quantity;
 int cost;
};

```

(6+6)

Assume that the file "iteminfo" already has 5 records. (L6)

a. Write a program that opens the file and appends the given new record to the file.

b. Display the content of the file.

4. a) Analyze and identify the situation when to use scripting language and system programming for application development. (L4) (3)

b) i) Discuss the features of markup languages in web Programming. (5) (L3)

ii) Explain the concept of static and dynamic libraries. (5)

Compare multi-paradigm languages with domain-specific languages, (L4)

- c) and also discuss when would you choose one over the other with typical use cases. (12)

**CO : COURSE OUTCOME; BTL : BLOOM'S TAXONOMY LEVEL**

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**FD/RSP**

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Roll No:

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PSG COLLEGE OF TECHNOLOGY, COIMBATORE 641 004

SEMESTER EXAMINATIONS, DECEMBER 2023

MCA Semester : 1

23MX12 STRUCTURED PROGRAMMING CONCEPTS

Time : 3 Hours

Maximum Marks : 100

**INSTRUCTIONS:**

1. Answer ALL questions. Each question carries 25 Marks.
2. Subdivision (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.

3. Course Outcome :  
Table

Qn.1 CO1.

Qn.2 CO2.

Qn.3 CO3.

Qn.4 CO4.

1. a) Compare and Contrast Compiler and Interpreter.  
b) Explain the essences of 'Structured Programming Concepts' and its availability in C with benefits in software applications development.  
c) i) Discuss the various criteria for classification of Programming Languages. Based on these criteria fit the position of Programming Language C in the classification chart. Mention when to use and when not use C for software development. [6]  
ii) Discuss the various properties of good programming language in the perspective of programmer and language developer. Enumerate reasons for the popularity of a programming language. Give simple illustrations in C for above said points [6]
2. a) Distinguish the uses of Iteration from Recursion in software applications development with an example.  
b) Compare and contrast the following programming language constructs: [3+4+3]  
i) Literals, Constants and Variables  
ii) Expressions and Statements  
iii) Sequence controls and Sequence points in Program execution  
c) i) 1) Write a C program to find all Saddle points of a given matrix along with their position details.
  - For the matrix M1, the (3, 1) cell is a saddle point because 7 is the smallest value in row 3 and the largest value in column 1.
  - For the matrix M2, the (2, 2) cell is a saddle point because 5 is the smallest value in row 2 and the largest value in column 2.
  - The matrix M3 does not have a saddle point. No value is both the smallest in its row and largest in its column.

| M1 |   |   |
|----|---|---|
| 1  | 2 | 3 |
| 4  | 5 | 6 |
| 7  | 8 | 9 |

| M2 |   |   |
|----|---|---|
| 9  | 1 | 2 |
| 8  | 5 | 7 |
| 3  | 4 | 6 |

| M3 |   |   |
|----|---|---|
| 8  | 1 | 9 |
| 7  | 2 | 6 |
| 3  | 4 | 5 |

[6]

- 2) Write an in-place C program that accepts a two dimensional array A of integers represented as `int A[m][n]`, rotate it 90 degrees clockwise. (In-place means minimal extra memory to be used, i.e., doesn't make a new array to copy into) [6]

Input

|    |    |    |    |
|----|----|----|----|
| 1  | 2  | 3  | 4  |
| 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

Output

|    |    |   |   |
|----|----|---|---|
| 13 | 9  | 5 | 1 |
| 14 | 10 | 6 | 2 |
| 15 | 11 | 7 | 3 |
| 16 | 12 | 8 | 4 |

[OR]

- ii) 1) Suppose in a C program, three strings PSG, Coimbatore, and Programming have to be stored. Calculate the memory requirement in bytes for each of the following two options of storing the strings in a computer with 64-bit addresses: [2+3]

Option-1: a statically declared 2D array of characters with 10 rows and 20 columns, with each word starting at a new row of the 2D array.

Option-2: a dynamically declared array of length three, where an array entry is a character pointer containing the starting address of a string.

- 2) Fill in the blanks for the following C program that reads in the information about a group of students from the user, and then prints the longest name among the group.

```
#include <stdio.h>
#include <string.h>
#define CLASS_SIZE 10
#define MAX_LEN 20

typedef struct
{
 char name[MAX_LEN];
 int age;
 float height; // in cm
} student;

_____ findlongestname(_____ class[]) // 2 marks
{
 int i, maxlen = 0, currlen, index;
 for (i = 0; i < CLASS_SIZE; i++)
 {
 if ((currlen = strlen(_____)) > maxlen) // 1 mark
 {
 maxlen = currlen; index = _____; // 1 mark
 }
 }
 return _____; // 1 mark
}

int main()
{
```

```

student class[CLASS_SIZE];
int i, index;
for (i = 0; i < CLASS_SIZE; i++)
{
 printf("\nEnter name: ");
 scanf("%s", class[i].name);
 printf("\nEnter age: ");
 scanf("%d", &class[i].age);
 printf("\nEnter height: ");
 scanf("%f", &class[i].height);
}
index = findlongestname(class);
printf("\nThe longest name is: %s", class[index].name);
return 0;
}

```

3. a) What is the output of the following C code snippet?

```

#include<stdio.h>
#include<stdio.h>
int main()
{
 char *arr[] = {"ant", "bat", "cat", "dog", "egg", "fly"};
 char *ptr1, *ptr2;
 int i;
 unsigned int k;
 ptr1 = (arr+1)[2];
 ptr2 = (*arr+2);
 for(i=0; i<10; i++)
 {
 k = k + (ptr2 - ptr1);
 k = k%4;
 }
 printf("Out1: %s || Out2: %s || Out3: %s\n", ptr1, ptr2, arr[k]);
 return 0;
}

```

- b) i) In this C declaration "long \*\*(\*(\*z)(void))[7](void);", What is z? Give syntax for the use of 'z'. [3]
- ii) You are given an  $m \times n$  array A of integers, each column of which is a sorted list of size m. Your task is to merge the n sorted lists and store the merged list in a one-dimensional array B. It is given that each column does not contain repetition(s) of integers, that is, the m integers in each sorted list are distinct from one another. However, integers may be repeated in different rows. During the merging step, you must remove all these repetitions. Use dynamic memory allocated arrays and

pointer arithmetic operations for processing. Print the arrays before and after merging operation. [7]

Input A    1      2      3      4  
             3      5      6      7  
             5      7      8      9

Output B    1      3      5      2      7      6      8      4      9

c) i) Discuss the following aspects of structured programming with illustrations from C.

a. Call by Value and Call by Reference [4]

b. Types of Storage Classes [5]

c. Function Pointers and its uses [3]

[OR]

ii) 1) Give the uses of 'extern', 'volatile' and 'const' keywords [4]

2) Given an array, the reverse Array function tries to reverse it using pointers. Fill up the blanks in the below code to achieve the functionality.

Example:

Input Array is {-1, 5, -8, -4, 5, 3}

Output is {3, 5, -4, -8, 5, -1} [8]

```
#include<stdio.h>
#include<stdio.h>
// Function to swap two memory contents

void swap(int *a, int *b)
{
 int temp = *a;
 _____ (1)
 *b = temp;
}

void reverseArray(int array[], int array_size)
{
 // leftPointer pointing at the beginning of the array
 int *leftPointer = _____ (2) ;

 // rightPointer pointing at end of the array
 int *rightPointer = _____ (3) ;

 while (_____(4))
 {
 swap(leftPointer, rightPointer);
 _____ (5) ;
 _____ (6) ;
 }
}
```

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4. a) How to decide when to use and when not use Patterns and Frameworks for application development?
- b) Compare and Contrast the features of System programming Languages and scripting languages in detail. Mention when to use each type of Programming Language.
- c) i) Explain the features offered by Static library, Dynamic library and Low-level Programming for application development in C. [8]
- ii) Discuss the role of GIT like tools in Source code Management aspects of any Web Application. [4]

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FD/RL